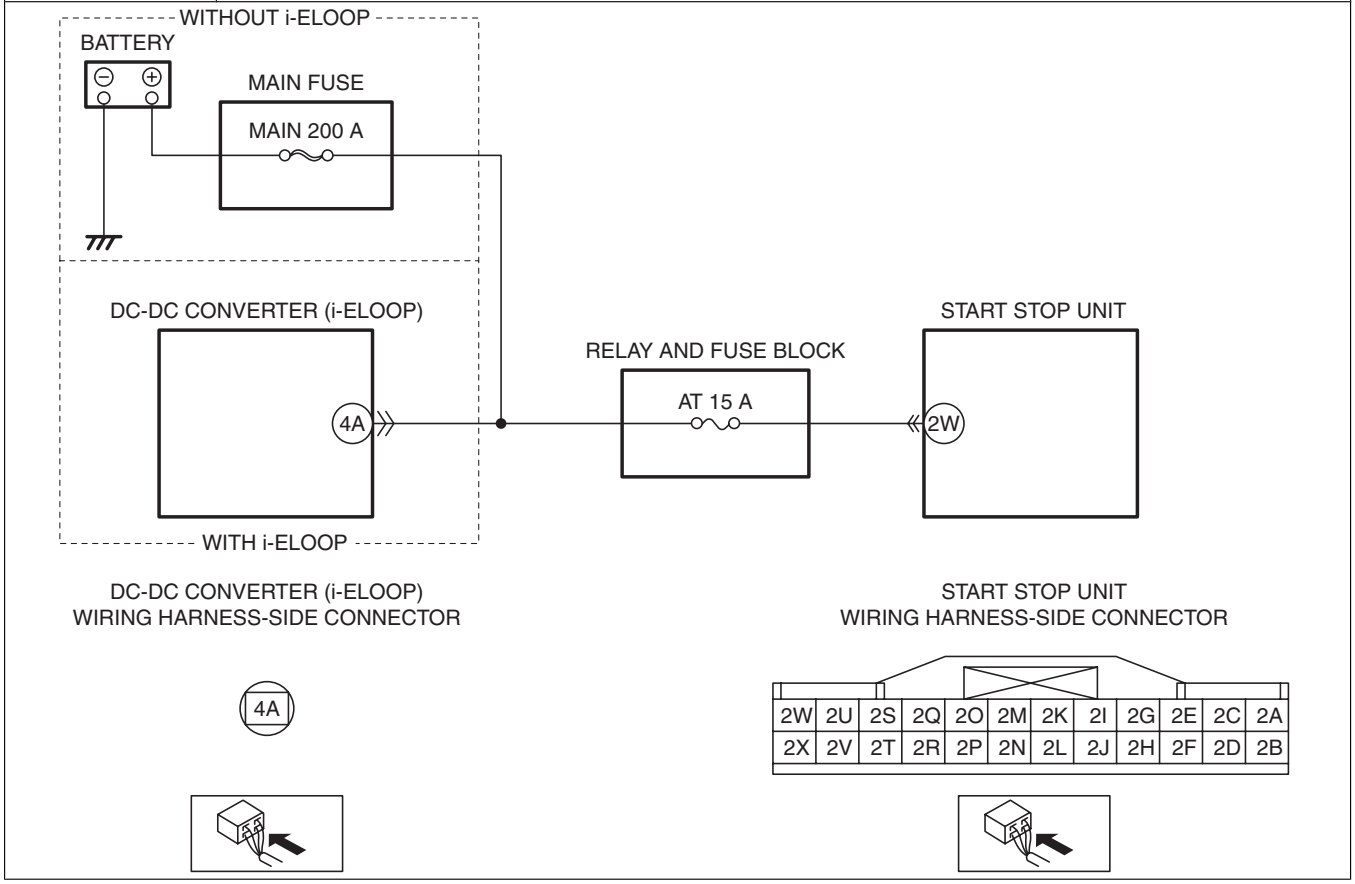
DTC P1794:16 [START STOP UNIT]

id0902p6023700

Description	Start stop unit power supply voltage (terminal 2W) low input
Detection condition	• Start stop unit power supply circuit voltage (terminal 2W) of **less than 6.5 V** is detected for **5 s or more**.
Fail-safe	Not applicable
Possible cause	• DTCs are stored in the PCM. • DTCs are stored in the DC-DC converter (i-ELOOP). — DC-DC converter (i-ELOOP) connector or terminal malfunction — DC-DC converter (i-ELOOP) malfunction • Battery malfunction • Generator malfunction • Start stop unit connector or terminal malfunction • Start stop unit power supply circuit (terminal 2W) malfunction — Without i-ELOOP: • Short to ground in wiring harness between MAIN 200 A fuse and start stop unit terminal 2W • AT 15 A fuse malfunction • MAIN 200 A fuse malfunction • Open circuit in wiring harness between battery positive terminal and start stop unit terminal 2W — With i-ELOOP: • Short to ground in wiring harness between AT 15 A fuse and start stop unit terminal 2W • AT 15 A fuse malfunction • Open circuit in wiring harness between DC-DC converter (i-ELOOP) terminal 4A and start stop unit terminal 2W • Start stop unit malfunction



Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY PCM DTCs** • Retrieve the PCM DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC TABLE [SKYACTIV-D 1.5].)
		No	**Without i-ELOOP:** • Go to Step 3. **With i-ELOOP:** • Go to the next step.
2	**VERIFY DC-DC CONVERTER (i-ELOOP) DTCs** • Retrieve the DC-DC converter (i-ELOOP) DTCs using the M-MDS. (See DTC INSPECTION [i-ELOOP].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [i-ELOOP].)
		No	Go to the next step.
3	**INSPECT BATTERY** • Inspect the battery. (See BATTERY INSPECTION.) • Is the battery normal?	Yes	Go to the next step.
		No	Recharge or replace the battery, then go to Step 8. (See BATTERY RECHARGING.) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
4	**INSPECT GENERATOR** • Inspect the generator. (See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].) (See GENERATOR INSPECTION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR INSPECTION [WITH i-ELOOP (SKYACTIV-D 1.5)].) • Is the generator normal?	Yes	Go to the next step.
		No	Replace the generator, then go to Step 8. (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].) (See GENERATOR REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
5	**INSPECT START STOP UNIT CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the start stop unit connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	**Without i-ELOOP:** • Go to the next step. **With i-ELOOP:** • Go to Step 7.
		No	Repair or replace the connector, then go to Step 8.

Step	Inspection		Action
6	**VERIFY START STOP UNIT POWER SUPPLY VOLTAGE (TERMINAL 2W)** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Display PID VPWR_B2 using the M-MDS. (See PID/DATA MONITOR INSPECTION [START STOP UNIT].) • Is the voltage **B+**?	Yes	Go to Step 8.
		No	Inspect the AT 15 A fuse and MAIN 200 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between MAIN 200 A fuse and start stop unit terminal 2W. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and start stop unit terminal 2W. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 8.

Step	Inspection		Action
7	**VERIFY START STOP UNIT POWER SUPPLY VOLTAGE (TERMINAL 2W)** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Display PID VPWR_B2 using the M-MDS. (See PID/DATA MONITOR INSPECTION [START STOP UNIT].) • Is the voltage **B+**?	Yes	Go to the next step.
		No	Disconnect the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) Inspect the AT 15 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between MAIN 200 A fuse and start stop unit terminal 2W. **If there is a common connector:** • Determine the malfunctioning location by inspecting the common connector and terminals for corrosion, damage, or disconnection and by inspecting the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground, or the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter (i-ELOOP) terminal 4A and start stop unit terminal 2W. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Connect the service plug, then go to the next step. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].)
8	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the start stop unit using the M-MDS. (See CLEARING DTC [START STOP UNIT].) • Switch the ignition ON (engine off or on) and wait for **5 s or more**. • Retrieve the start stop unit DTCs using the M-MDS. (See DTC INSPECTION [START STOP UNIT].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the start stop unit. (See START STOP UNIT REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
9	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [START STOP UNIT].)
		No	DTC troubleshooting completed.